

Computational Reproduction and Extension of a Five-Ocean Carbon Sink Box Model

Reference: Sunny, E. M., Ashok, B., Balakrishnan, J., & Kurths, J. (2023). *The ocean carbon sinks and climate change*. *Chaos*, 33, 103134.

Abstract

This report describes a computational implementation of the five-ocean conceptual box model introduced by Sunny et al. (2023), together with two neural-network-based surrogates of the same dynamics and a reconstruction of the paper’s spatial carbon sink/source maps (Figure 7). The mechanistic model reproduces the coupled ordinary differential equations governing $\text{CO}_2(\text{aq})$ concentration, a reaction-product concentration, and salinity for the North Atlantic, South Atlantic, Southern, Pacific, and Indian Oceans, driven by the observed Mauna Loa atmospheric CO_2 record rather than the paper’s linear Keeling-curve approximation. A physics-informed neural network (PINN) and a Neural ODE were each trained to approximate the system’s evolution from partial data, constrained by the governing equations. Finally, the paper’s spatial sink/source distributions were reconstructed using the box model’s ocean-average CO_2 values, real reanalysis wind fields, and an explicit land/sea mask. Descriptive statistics (mean, standard deviation, minimum, maximum) are reported for every stage of the analysis.

1. Introduction

Sunny et al. (2023) propose a five-box model of the global ocean to capture the temporal evolution of dissolved CO_2 and salinity in each major ocean basin, driven by surface and deep-water currents, freshwater fluxes, and rising atmospheric CO_2 . The present work reproduces that model numerically, extends its atmospheric forcing term using real observational data, evaluates whether machine-learning surrogates can approximate its dynamics, and reconstructs its spatial sink/source visualization, including the specific finite-difference calculation used to localize carbon sinks and sources within each basin.

2. Methods

2.1 Box Model Formulation

The system state consists of fifteen coupled variables: $\text{CO}_2(\text{aq})$ concentration (C), the reaction-product concentration (N), and salinity (S), each defined for the five ocean boxes (North Atlantic [NA], South Atlantic [SA], Southern Ocean [SO], Pacific [PO], Indian Ocean [IO]).

- **Atmospheric forcing.** Rather than the paper’s fitted linear relation $\text{pCO}_2(t) = 0.18152t + 367.41$, the implementation uses the observed Mauna Loa monthly

CO₂ record directly, interpolated at each integration timestep and linearly extrapolated beyond the end of the available record.

- **Geometry and initial conditions.** Ocean volumes and surface areas are taken from the paper’s Appendix A; initial CO₂ and salinity values for January 2000 are taken from Table I, with the initial reaction-product concentration derived via the hydration equilibrium constant ($K_{eq} = 0.0011$).
- **Sea-surface temperature.** Each ocean’s SST follows the linear-in-time fits reported in the paper’s Appendix D.
- **Air-sea exchange.** The air-sea CO₂ flux follows Eq. (5), using a temperature-dependent gas-transfer velocity (Schmidt-number relation and Wanninkhof parameterization, Appendix B) and a temperature- and salinity-dependent solubility constant K_0 (Weiss relation, Appendix B, Eq. B2).
- **Reversible carbonate chemistry.** Conversion between C and N uses the paper’s fixed forward/reverse rate constants ($K_c = 0.037 \text{ s}^{-1}$, $K_n = 23 \text{ s}^{-1}$).
- **Inter-ocean transport.** Deep, density-driven flows (fixed rate constant, 60 Sv) and wind-driven surface currents (10–60 Sv, per the paper’s cited literature values) move C, N, and S between basins according to the connectivity in the paper’s Figure 2, with deep-flow CO₂ transport using the density/temperature/salinity-corrected concentration (Eq. 10).
- **Salinity balance.** Salinity evolves per Eq. (4), incorporating precipitation-minus-evaporation freshwater flux, polar ice melt (NA and SO only), and Amazon River freshwater input (NA only, 0.24 Sv).
- **Numerical integration.** The full 15-dimensional system was integrated with an implicit stiff solver (Radau method, relative tolerance 10^{-6} , absolute tolerance 10^{-10}) over the span of the available Mauna Loa record.

2.2 Physics-Informed Neural Network Surrogate

A feed-forward network (two hidden layers, 64 units each, tanh activations) maps normalized time to all fifteen state variables simultaneously, with outputs bounded to physically plausible ranges via scaled sigmoids. The network is optimized against a data term (CO₂ concentration only, over an early window of months 0–180 of ~317 available) combined with a physics term requiring the automatic-differentiation-derived time derivative of the network’s own output to match the governing equations evaluated at that same predicted state, sampled at collocation points spanning the full time range. Per-channel time derivatives were computed via an explicit channel-by-channel Jacobian rather than a summed gradient, and the network predicts the full 15-dimensional state itself, avoiding both an incorrect physics residual and data leakage from the reference solution.

2.3 Neural ODE Surrogate

A second surrogate parameterizes the governing equations’ right-hand side directly: a small network (two hidden layers, 64 units, tanh) maps the current normalized 15-dimensional state plus the current normalized atmospheric CO₂ forcing to an instantaneous rate of change, integrated forward in time with a fixed-step RK4 solver (`torchdiffeq`). State variables were normalized using training-window mean and standard deviation prior to being passed through the network. The network is optimized

to minimize the mean-squared error between the integrated trajectory and the true box-model trajectory over the training window.

2.4 Data Source Verification

Prior to reconstructing spatial maps, NOAA’s ERDDAP data-server search API was queried to confirm which gridded wind/ocean datasets remain actively hosted, since the original data sources cited in the 2023 paper may no longer be maintained years later.

2.5 Spatial Reconstruction of Carbon Sink/Source Distributions

This is the component of the work that localizes carbon sinks and sources *within* each ocean basin, reproducing the paper’s Figure 7. It combines each ocean’s box-model average CO₂ concentration with local wind fields through a convective-derivative calculation (paper Eqs. 11–13):

1. The box model was re-solved to obtain each ocean’s average CO₂ concentration at the specific months analyzed in the paper (January 2003, 2005, 2007, 2011), which serve as the boundary condition for the spatial calculation in each basin.
2. The globe was discretized onto a 2°×2° latitude/longitude grid, with each ocean cell assigned to one of the five basins using the paper’s stated geographic boundaries (70°W, 20°E, 140°E, 60°S), and land cells excluded via an independent global land/sea mask dataset.
3. Real reanalysis wind fields were retrieved for January of each target year from a NOAA ERDDAP-hosted geostrophic wind product (no synthetic wind data).
4. The finite-difference form of the convective-derivative equation was solved locally at each grid cell, using wind components as advection velocities and the paper’s own published dC/dt (“f”) values (Appendix C) as the source/sink forcing term — where wind is strong, the local CO₂ anomaly relative to the ocean average is spread thin; where wind is weak, the same forcing term produces a larger local anomaly.
5. The resulting field was rendered with discrete color-bin boundaries matching the paper’s own published Figure 7 legend, across a panel layout mirroring the paper’s (a)–(e) structure.

Two methodological refinements were incorporated into this calculation. First, wind-speed thresholding was applied to the resultant wind magnitude rather than to its east-west and north-south components independently, preserving the true wind direction in low-wind-speed regions. Second, the geostrophic wind formulation is not defined in the immediate vicinity of the equator, since its derivation depends on the Coriolis parameter, which vanishes there; a physically motivated fallback value was applied in this band to maintain continuous spatial coverage. A bounding constraint was also applied to the computed field to maintain physically realistic values under low-wind-speed conditions.

3. Results

3.1 Box Model Trajectories

The system was integrated over 317 months (January 2000 onward).

Table 1. CO₂(aq) concentration statistics per ocean, full 317-month trajectory (mol/m³)

Ocean	Mean	SD	Min	Max	Final value
North Atlantic	0.013495	0.000265	0.013188	0.014130	0.014037
South Atlantic	0.013581	0.000576	0.013088	0.015790	0.013877
Southern Ocean	0.019683	0.002086	0.014000	0.022301	0.022301
Pacific	0.014219	0.001181	0.013400	0.018340	0.013854
Indian Ocean	0.015305	0.000422	0.014839	0.016710	0.015949

Table 2. Salinity statistics per ocean, full 317-month trajectory (psu)

Ocean	Mean	SD	Min	Max	Final value
North Atlantic	34.8407	0.0413	34.7695	34.9120	34.7695
South Atlantic	35.3871	0.0277	35.3392	35.4350	35.3392
Southern Ocean	34.4695	0.0246	34.4270	34.5121	34.5121
Pacific	34.6667	0.0008	34.6654	34.6680	34.6654
Indian Ocean	34.4084	0.0751	34.2787	34.5380	34.2787

The Southern Ocean has both the highest mean CO₂(aq) concentration and the largest standard deviation of the five basins, consistent with the paper's finding of the Southern Ocean as the dominant and most dynamically active carbon sink. Salinity standard deviations are small relative to the reference 35 psu value across all basins, with the Pacific essentially static (SD = 0.0008) and the Indian Ocean showing the most salinity variability (SD = 0.0751).

Table 3. Box-averaged CO₂(aq) concentration at the four spatial-reconstruction target months (mol/m³)

Month	NA	SA	SO	PO	IO	Cross-ocean mean	Cross-ocean SD
Jan 2003	0.012937	0.014219	0.016619	0.016048	0.015497	0.015064	0.001328
Jan 2005	0.012547	0.013641	0.017767	0.015112	0.015102	0.014834	0.001755

Month	NA	SA	SO	PO	IO	Cross-ocean mean	Cross-ocean SD
Jan 2007	0.012345	0.013298	0.018616	0.014490	0.014904	0.014731	0.002141
Jan 2011	0.012255	0.013036	0.019778	0.013845	0.014846	0.014752	0.002656

Table 4. Per-ocean statistics across the four target months

Ocean	Mean	SD	Min	Max
North Atlantic	0.012521	0.000262	0.012255	0.012937
South Atlantic	0.013548	0.000443	0.013036	0.014219
Southern Ocean	0.018195	0.001157	0.016619	0.019778
Pacific	0.014874	0.000813	0.013845	0.016048
Indian Ocean	0.015087	0.000255	0.014846	0.015497

The cross-ocean standard deviation increases monotonically across the four target years ($0.001328 \rightarrow 0.001755 \rightarrow 0.002141 \rightarrow 0.002656$), indicating that the five ocean basins diverge from one another over time rather than converging — driven primarily by the Southern Ocean’s steady rise ($0.016619 \rightarrow 0.019778$, a 19.0% increase) against the North Atlantic and South Atlantic’s steady decline.

3.2 Physics-Informed Neural Network Results

Table 5. PINN predicted $\text{CO}_2(\text{aq})$ concentration statistics, full 317-month trajectory (mol/m^3)

Ocean	Mean	SD	Min	Max
North Atlantic	0.014638	0.000001	0.014637	0.014639
South Atlantic	0.014880	0.000001	0.014879	0.014881
Southern Ocean	0.018948	0.000000	0.018948	0.018949
Pacific	0.015779	0.000000	0.015778	0.015779
Indian Ocean	0.016188	0.000000	0.016187	0.016189

The PINN’s per-ocean standard deviations are three to four orders of magnitude smaller than the box model’s (Table 1), indicating that the network converges to an almost constant value per ocean rather than reproducing the box model’s temporal variation, while preserving the correct relative ordering between oceans (Southern Ocean highest, North Atlantic lowest).

3.3 Neural ODE Results

Table 6. Neural ODE predicted CO₂(aq) concentration statistics, full trajectory (mol/m³)

Ocean	Mean	SD	Min	Max
North Atlantic	0.013462	0.000232	0.013192	0.014130
South Atlantic	0.013524	0.000580	0.013095	0.015790
Southern Ocean	0.019745	0.002158	0.014000	0.022793
Pacific	0.014078	0.001267	0.013091	0.018340
Indian Ocean	0.015242	0.000381	0.014846	0.016710

Unlike the PINN, the Neural ODE’s per-ocean means and standard deviations are close in magnitude to the box model’s own statistics (Table 1) — e.g., Southern Ocean mean 0.019745 vs. 0.019683, SD 0.002158 vs. 0.002086 — indicating that the Neural ODE captures substantially more of the box model’s temporal variability than the PINN.

3.4 Spatial Sink/Source Reconstruction Results

Table 7. Grid cell distribution (2°×2° global grid, land/ocean assignment)

Region	Cell count	Percentage of grid
Land	5,446	33.6%
North Atlantic	1,378	8.5%
South Atlantic	1,030	6.4%
Southern Ocean	1,129	7.0%
Pacific	4,621	28.5%
Indian Ocean	2,596	16.0%
Total	16,200	100%

Table 8. Sink/source field value range by target year (mol/m³)

Year	Lower bound	Upper bound
2003	−0.7619	2.0166
2005	0.0189	2.0178
2007	0.0577	2.0186
2011	0.0484	2.0198
Mean across years	−0.1592	2.0182
SD across years	0.3482	0.0012

The upper field bound is essentially constant across all four target years (SD = 0.0012), while the lower bound is more variable (SD = 0.3482) and driven largely by 2003, whose lower bound (−0.7619) is well below the other three years’ lower

bounds (0.0189, 0.0577, 0.0484) — indicating a localized, strong carbon-source signal specific to 2003 rather than a broad shift in the whole field.

Table 9. Wind-field data gaps addressed via the fallback procedure, per affected ocean box (representative target year)

Statistic	Value
Mean	346.0 cells
SD	234.5 cells
Min	124 cells
Max	692 cells
Total (summed across affected boxes)	1,384 cells

These data gaps are concentrated near the equator, consistent with the known limitation of the geostrophic wind product used, whose underlying formulation is not defined where the Coriolis parameter approaches zero. The applied fallback procedure ensured continuous spatial coverage across each ocean basin, and the resulting five-panel figure (matching the paper’s Figure 7 a–e layout and discrete color legend) was generated successfully for all target years. The Miocene-era alternate-Amazon-flow scenario corresponding to the paper’s Figure 7(f) was not computed and is represented as a placeholder panel.